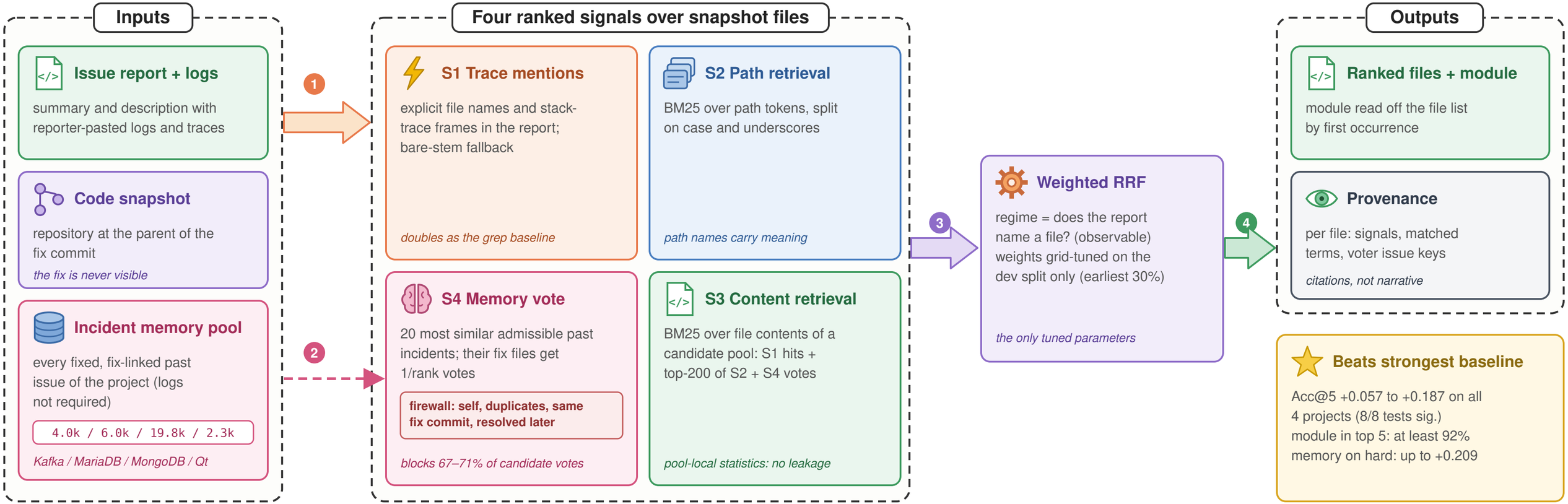
Inputs
Issue report + logs
summary and description with reporter-pasted logs and traces
Code snapshot
repository at the parent of the fix commit
the fix is never visible
Incident memory pool
every fixed, fix-linked past issue of the project (logs not required)
4.0k / 6.0k / 19.8k / 2.3k
Kafka / MariaDB / MongoDB / Qt
1
2
Four ranked signals over snapshot files
S1 Trace mentions
explicit file names and stack-trace frames in the report; bare-stem fallback
doubles as the grep baseline
S2 Path retrieval
BM25 over path tokens, split on case and underscores
path names carry meaning
S4 Memory vote
20 most similar admissible past incidents; their fix files get 1/rank votes
firewall: self, duplicates, same fix commit, resolved later
blocks 67–71% of candidate votes
S3 Content retrieval
BM25 over file contents of a candidate pool: S1 hits + top-200 of S2 + S4 votes
pool-local statistics: no leakage
3
Weighted RRF
regime = does the report name a file? (observable)
weights grid-tuned on the dev split only (earliest 30%)
the only tuned parameters
4
Outputs
Ranked files + module
module read off the file list by first occurrence
Provenance
per file: signals, matched terms, voter issue keys
citations, not narrative
Beats strongest baseline
Acc@5 +0.057 to +0.187 on all 4 projects (8/8 tests sig.)
module in top 5: at least 92%
memory on hard: up to +0.209